Fact-Checking Climate Claims with Sparse Retrieval and Cross-Encoder Reranking

Group 56

Abstract

We describe a fact-checking system for climate-science claims that retrieves supporting evidence from a 1.2M-passage corpus and classifies each claim as SUPPORTS, REFUTES, NOT ENOUGH INFO, or DISPUTED. The system is built as a modular retrieval-then-classification pipeline: BM25 selects a broad candidate pool, a MiniLM cross-encoder reranks claim-evidence pairs, a relative-delta rule selects a variable number of final evidence passages, addressing the issue that cross-encoder logits are typically uncalibrated across different claims. A second MiniLM cross-encoder predicts the claim label from the claim and selected evidence. We tune the pipeline with a sequential greedy ablation rather than a full factorial search, using recall@500 for first-stage retrieval to define the theoretical performance upper bound, evidence F-score for selection, and accuracy plus per-class recall for classification. On the development set, the final system obtains evidence F-score 0.2448, label accuracy 0.4805, and harmonic mean 0.3243. The largest gain comes from the retrieval pipeline: final retrieval doubles evidence F-score over raw BM25 top-5 evidence. Classification remains the main weakness, with a strong bias toward SUPPORTS and zero recall for REFUTES.

1 Introduction

Automated fact-checking requires both finding relevant evidence and reasoning over it. This coupling is central to FEVER-style verification (Thorne et al., 2018) and is especially important for climate claims, where statements are often paraphrastic, evidence may be multi-sentence, and labels can depend on uncertainty or disagreement rather than simple true/false judgments (Diggelmann et al., 2020). In this assignment, each system receives a claim and must return evidence passage identifiers from a corpus of 1,208,827 passages, together with one of four labels: SUPPORTS, REFUTES, NOT ENOUGH INFO (NEI), or DISPUTED.

The main challenge is that the final score rewards balanced performance. A strong classifier cannot help if the right evidence never enters the candidate pool, while high-recall retrieval is insufficient if the selected passages are noisy or the classifier falls back to majority-class behavior. Our design therefore separates the problem into three questions: can sparse retrieval include the gold evidence in a manageable pool, can a cross-encoder reranker improve the final evidence set, and can a lightweight classifier use that evidence to improve over a majority-label baseline?

We find that the retrieval side is effective but still far from solved. Stemming and BM25 tuning improve recall@500, and MiniLM reranking plus relative-delta evidence selection raises development evidence F-score from 0.1234 for raw BM25 top-5 to 0.2448. The classifier, however, improves accuracy over an all-SUPPORTS baseline by only 3.9 percentage points. Error analysis shows that retrieval failures and classification failures are both substantial, and oracle retrieval does not automatically improve label accuracy, suggesting that evidence reasoning remains fragile and heavily dependent on specific data distributions rather than robust logic.

2 Approach

2.1 Task Formulation

For each claim c, the system ranks evidence passages $e \in E$ and returns a selected evidence set $R(c)$ plus a label $\hat{y} \in Y$, where $Y = \{\text{SUPPORTS}, \text{REFUTES}, \text{NEI}, \text{DISPUTED}\}$. The official score reports evidence retrieval F-score F, label accuracy A, and their harmonic mean:

$$H = \frac{2FA}{F + A}. \quad (1)$$

We optimize modules with local development metrics, then evaluate the full pipeline with F, A, and H. This is a sequential greedy ablation: each stage is chosen conditional on earlier stages. This is less exhaustive than a full factorial search, but it keeps training feasible under the assignment's free-Colab constraint.

2.2 First-Stage Retrieval

The first stage uses sparse lexical retrieval. We compare TF-IDF and BM25 (Robertson and Zaragoza, 2009), with three preprocessing settings: lowercasing only, lowercasing plus stopword removal, and lowercasing plus stopword removal plus Porter stemming (Porter, 1980). We choose the first-stage configuration by development recall@500, because evidence that is absent from the top-500 pool cannot be recovered by downstream reranking.

The selected setting is BM25 with lowercasing, stopword removal, Porter stemming, $k_1 = 1.2$, and $b = 0.5$. This retrieves 500 candidates per claim for train, development, and test splits.

2.3 Cross-Encoder Reranking

The reranker is cross-encoder/ms-marco-MiniLM-L-12-v2, a MiniLM model (Wang et al., 2020) trained for MS MARCO passage ranking (Bajaj et al., 2016). Unlike a bi-encoder, the cross-encoder reads the claim and evidence passage jointly and outputs a relevance logit $s(c, e)$.

Training pairs are built from the selected BM25 top-500 pool. Gold claim-evidence pairs are positives; negatives are sampled from BM25 candidates that are not gold evidence. This produces 20,610 training pairs, with 4,122 positives and four hard negatives per positive. We train for up to three epochs and select the checkpoint with the best development evidence F-score under a quick relative-delta check. The first epoch is best; later epochs reduce development F-score, consistent with overfitting to the small pair dataset. Per-epoch numbers are reported in Section 4.3.

2.4 Evidence Selection

Reranking gives an ordered list and logits, but the submission must return a small final evidence set. We compare fixed-K, global-threshold, and relative-delta selection. The final system uses relative-delta selection:

$$R(c) = \{e_i : s(c, e_i) \geq \max_j s(c, e_j) - \delta\}. \quad (2)$$

This rule adapts to each claim's score scale instead of assuming that logits are calibrated across claims. The selected threshold is $\delta = 1.5$, which yields 4.05 selected evidence passages per claim on average.

Item	Count
Train claims	1,228
Dev claims	154
Test claims	153
Evidence passages	1,208,827
Mean gold evidence / claim	3.357

Table 1: Dataset summary.

2.5 Claim Classification

The classifier receives the claim concatenated with up to five selected evidence passages and predicts one of the four labels. During classifier training, the evidence input is the union of gold and retrieved evidence, deduplicated and truncated to five passages; at development and test time, only retrieved evidence is available. We compare distilbert-base-uncased (Sanh et al., 2019), cross-encoder/ms-marco-MiniLM-L-12-v2, and cross-encoder/nli-MiniLM2-L6-H768. Models are trained with learning rate 10^{-5} for up to four epochs, batch size 16 for training and 32 for evaluation.

Class imbalance is severe: SUPPORTS is the largest class and DISPUTED the smallest. We therefore compare no class weighting, square-root inverse-frequency weighting, and full inverse-frequency weighting. The final submitted classifier uses MiniLM-MARCO without class weighting, because stronger weighting increases minority-class pressure but sharply damages overall accuracy.

3 Experiments

3.1 Data and Evaluation

The dataset statistics are shown in Table 1. Training labels are imbalanced: 42.26% SUPPORTS, 16.21% REFUTES, 31.43% NEI, and 10.10% DISPUTED. The average claim has 3.357 gold evidence passages, so selecting only one passage is usually insufficient.

The official evaluation script computes aggregate evidence F-score by comparing predicted evidence identifiers to gold identifiers, classification accuracy by comparing predicted labels to gold labels, and H from Equation 1. We report development results because test labels are unavailable locally; the notebook also generates test-output.json for leaderboard submission.

3.2 Experimental Setup

All reported development numbers are produced by v13.ipynb and summarised in v13_experiment_summary.md. We use seed 42 for the main run. Reranker and classifier checkpoints are saved under outputs_notebook_v13; candidate lists, cross-encoder scores, and final evidence sets are cached for reproducibility. The full search is intentionally sequential: retraining rerankers and classifiers for every preprocessing and BM25 combination would multiply the compute cost.

3.3 Training, Search, and Class Balance

To reduce overfitting, both neural stages use validation-set checkpoint selection. The reranker is trained for three epochs, but after each epoch we rescore the development candidate pool and keep the checkpoint with the highest development retrieval F-score. The classifier is trained for up to four epochs, and the saved checkpoint is the epoch with the best development accuracy, using harmonic mean as a tie-breaker. In practice, this behaves as early stopping because the final model is not necessarily the last epoch. For example, the reranker loss keeps decreasing from 0.3809 to 0.2291, but development F-score decreases after epoch 1; therefore epoch 1 is selected.

Table 2 summarises the main search space. We use grid search for the stages where the choice can be evaluated directly on development data: first-stage retrieval is selected by recall@500, evidence selection is selected by evidence F-score, and classifier settings are selected by development accuracy and per-class behaviour. Class imbalance is handled through weighted cross-entropy. For a class y with training count n_y, the unnormalised weight is $(N/n_y)^p$, where $p = 0$ gives no weighting, $p = 0.5$ gives square-root inverse-frequency weighting, and $p = 1$ gives full inverse-frequency weighting. The weights are then renormalised so that their sum equals the number of labels.

3.4 Baselines

We use two majority-label baselines. Both predict SUPPORTS for every claim, matching the largest development label prior. One uses raw BM25 top-5 evidence, and the other uses the final retrieval output. This isolates whether improvements come from evidence retrieval or claim classification.

Stage	Grid or selection rule
Sparse retrieval	3 preprocessing $\times$ 5 retrieval configs
BM25s top-K diagnostic	$K \in \{5, 20, 50, 100, 200, 500\}$
Reranker checkpoint	best dev F over 3 epochs
Evidence selection	fixed-K, threshold, relative-δ
Classifier backbone	DistilBERT, MiniLM-MARCO, MiniLM-NLI
Class weighting	$p \in \{0, 0.5, 1\}$

Table 2: Hyperparameter search and checkpoint selection used in the experiments.

BM25s top-K	Recall	Precision	F1
5	0.1603	0.0896	0.1077
20	0.2732	0.0390	0.0662
50	0.3761	0.0223	0.0416
100	0.4392	0.0131	0.0252
200	0.5173	0.0078	0.0154
500	0.6179	0.0038	0.0075

Table 3: BM25s coarse-ranking diagnostic on the development set with different candidate-pool sizes.

4 Results

4.1 BM25s Candidate Pool Size

Before applying the neural reranker, we first examine how many BM25s candidates should be kept from the coarse ranking stage. Table 3 reports a development-set diagnostic using an earlier default BM25 setting. As K increases, recall rises from 0.1603 at top-5 to 0.6179 at top-500, showing that many gold evidence passages are not ranked in the very top positions by lexical matching alone. At the same time, precision and evidence F1 decrease because the predicted set becomes much larger and noisier. We therefore do not use top-500 as the final evidence output. Instead, we use it as a candidate pool for later reranking and selection.

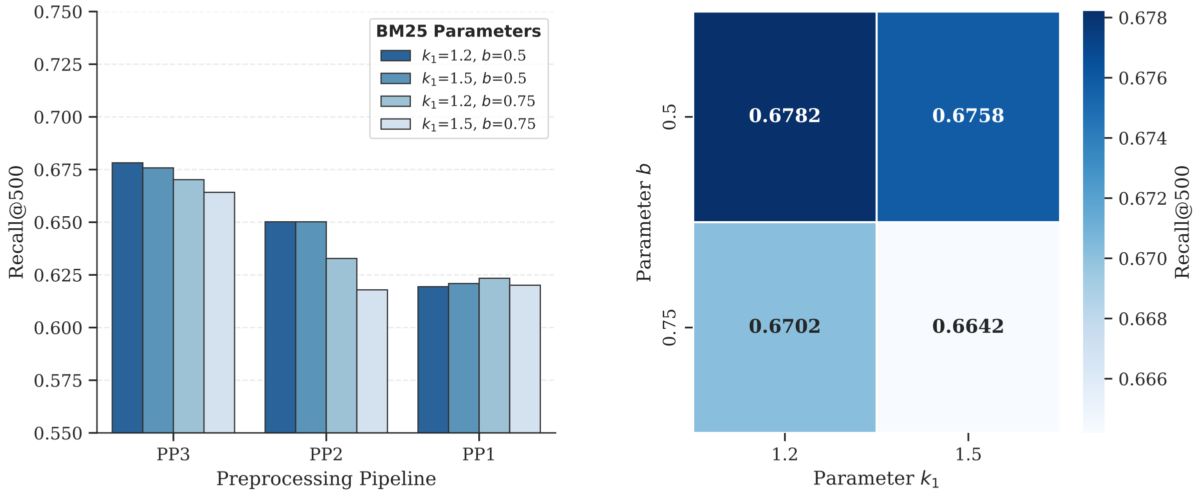
This comparison explains why we choose a large candidate pool for the first stage. A small K gives a cleaner but low-recall evidence list, while a large K gives the downstream cross-encoder a much better chance to see at least one gold passage. The top-500 setting is therefore a recall-oriented design choice rather than a final retrieval output.

4.2 Sparse Retrieval and Preprocessing Analysis

This stage conducts a grid search through 3 preprocessing schemes and 5 retrieval configurations, resulting in a total of 15 experimental groups. The experiment takes BM25 top-5 as the initial base-

ID	preproc	retrieval	method	dev_recall@500	dev_F@5	time_s	k1	b
0	PP3	R-both	bm25	0.678247	0.123418	21.4	1.2	0.50
1	PP3	R-b05	bm25	0.675758	0.121666	24.5	1.5	0.50
2	PP3	R-k12	bm25	0.670238	0.126551	26.1	1.2	0.75
3	PP3	R-default	bm25	0.664177	0.117692	24.7	1.5	0.75
4	PP3	R-TFIDF	tfidf	0.651299	0.091857	9.3	-	-
5	PP2	R-b05	bm25	0.650216	0.116687	23.4	1.5	0.50
6	PP2	R-both	bm25	0.650216	0.120908	23.6	1.2	0.50
7	PP2	R-k12	bm25	0.632792	0.113966	24.8	1.2	0.75
8	PP1	R-k12	bm25	0.623377	0.123413	28.2	1.2	0.75
9	PP1	R-b05	bm25	0.620887	0.125778	29.7	1.5	0.50
10	PP1	R-default	bm25	0.620130	0.107040	23.2	1.5	0.75
11	PP1	R-both	bm25	0.619372	0.131298	26.7	1.2	0.50
12	PP2	R-default	bm25	0.617857	0.107689	25.2	1.5	0.75
13	PP1	R-TFIDF	tfidf	0.612121	0.090012	12.2	-	-
14	PP2	R-TFIDF	tfidf	0.602597	0.086786	8.8	-	-

Table 4: Retrieval Performance Across Different Preprocessing and Parameter Configurations



(a) IMPACT OF PREPROCESSING

(b) HYPERPARAMETER SEARCH

Figure 1: Retrieval performance analysis. (a) Comparison of different preprocessing pipelines across various BM25 parameter configurations. (b) Grid search of hyperparameters k_1 and b using the optimal PP3 pipeline.

line, and Recall@500 is used as the primary metric to measure the theoretical upper limit of system performance.

As shown in Table 4, PP3 + BM25 ($k_1 = 1.2$, $b = 0.5$) achieved the best results. The performance of BM25 is consistently better than that of TF-IDF, and the recall rate after incorporating Porter Stemming (PP3) increased by approximately 2.8% compared to PP2. This confirms our expectation that given the specialized vocabulary in the climate domain, the normalization of word forms is crucial to capture semantic associations. In addition, by lowering the b parameter to 0.50, Recall was steadily improved, indicating that when dealing with the generally short sentences in the evidence corpus, a reduced length penalty can capture key information more accurately than the default configuration.

Although the best-configured Recall@500 reached 0.6782, the F1-score of its direct Top-5 was only 0.1234. Combined with the subsequent Error Bucket analysis, we found that 27.27% of claims still face Retrieval Failure. This demonstrates that sparse retrieval based on lexical matching has inherent limitations when dealing with a massive 1.2M-passage corpus; many climate statements involve complex semantic expressions, making it impossible to bridge the "semantic gap" by relying on keyword matching alone. These quantitative results strongly justify the necessity of introducing deep semantic reranking (Reranker) in the next stage to compensate for the precision deficit of lexical matching.

Strategy	Param.	Dev F
Relative-delta	1.50	0.2448
Relative-delta	1.00	0.2329
Relative-delta	2.00	0.2321
Fixed-K	3	0.2156
Fixed-K	4	0.2149
Threshold	0.90	0.2079

Table 5: Evidence selection results after reranking.

Backbone	A	Supp.	Ref.	Disp.
MiniLM-MARCO	0.4935	0.9706	0.0000	0.0000
MiniLM-NLI	0.4805	0.8088	0.3333	0.1111
DistilBERT	0.4481	0.8088	0.0370	0.0000

Table 6: Classifier backbone comparison. Columns after accuracy show per-class recall for selected classes.

4.3 Reranking and Evidence Selection

After selecting the sparse retriever, we train the MiniLM reranker on the best top-500 BM25 candidate pool. The reranker peaks after one epoch: training loss decreases from 0.3809 to 0.2632 and 0.2291 over three epochs, but quick development F drops from 0.1961 to 0.1911 and 0.1806. This suggests that the cross-encoder quickly learns useful relevance signals but begins to overfit the 20,610 pair training set.

Selection results in Table 5 show that relative-delta selection is better than fixed-K and global thresholds. A global threshold assumes scores are calibrated across claims, but cross-encoder logits vary by claim. The relative rule instead asks how close each evidence passage is to the best passage for the same claim.

4.4 Classification

Table 6 compares classifier backbones with square-root inverse-frequency weighting fixed for this stage. MiniLM-MARCO gives the highest overall accuracy, but the per-class recalls reveal a serious bias: it obtains 0.9706 recall for SUPPORTS, 0 for REFUTES, and 0 for DISPUTED. The NLI-oriented MiniLM is less accurate overall but recovers some REFUTES and DISPUTED examples, suggesting that the final classifier's accuracy is partly driven by label-prior behavior rather than robust evidence reasoning.

After selecting MiniLM-MARCO, we compare class weighting in Table 7. Full inverse-frequency weighting raises DISPUTED recall to 0.8889 but collapses accuracy to 0.2468. Square-root weighting gives the best H by a narrow margin, but predicts no DISPUTED claims correctly. We choose no weighting for the final system because it preserves some DISPUTED recall while avoiding the large accuracy collapse.

Weighting	A	H	Ref.	Disp.
None	0.4805	0.3243	0.0000	0.2778
Sqrt inv. freq.	0.4870	0.3258	0.0000	0.0000
Inv. freq.	0.2468	0.2458	0.0741	0.8889

Table 7: Class weighting ablation for the MiniLM-MARCO classifier.

System	F	A	H
All SUPPORTS + BM25 top-5	0.1234	0.4416	0.1929
All SUPPORTS + final retrieval	0.2448	0.4416	0.3150
Final system	0.2448	0.4805	0.3243

Table 8: Main development-set results.

4.5 Final Development Results

Table 8 compares the final system to majority-label baselines. The final retrieval output doubles evidence F-score over BM25 top-5 evidence, improving H from 0.1929 to 0.3150 even when the label is always SUPPORTS. The classifier then raises accuracy from 0.4416 to 0.4805, giving final $H = 0.3243$. Thus most of the measured improvement comes from retrieval, while classification contributes a smaller but real gain.

The confusion matrix in Table 9 makes the classifier limitation clear. The system predicts SUPPORTS for most claims and never predicts REFUTES correctly. This is the main reason accuracy remains close to the majority baseline despite better evidence retrieval.

4.6 Error Analysis

Seed variance confirms that the classifier result should not be overinterpreted. With the same retrieval fixed and seeds 42, 123, and 2024, mean accuracy is 0.4892 ± 0.0201 and mean H is 0.3262 ± 0.0046. Retrieval F-score is unchanged because evidence selection is fixed.

Because DISPUTED claims often require detecting conflict between evidence passages, we also tried an open-source Qwen2.5-3B-Instruct cascade. For claims currently predicted as SUPPORTS or REFUTES with at least two retrieved passages, the model was prompted to answer whether the passages contradicted each other; a positive answer would change the label to DISPUTED. As Table 10 shows, the cascade made no label changes on the

Gold	Sup.	Ref.	NEI	Disp.
SUPPORTS	65	0	2	1
REFUTES	24	0	3	0
NEI	37	0	4	0
DISPUTED	13	0	0	5

Table 9: Final classifier confusion matrix on the development set. Rows are gold labels and columns are predictions.

Setting	F	A	H
Final system	0.2448	0.4805	0.3243
LLM cascade	0.2448	0.4805	0.3243

Table 10: Qwen2.5-3B-Instruct DISPUTED cascade on 130 eligible development claims. The cascade flipped zero labels.

130 eligible development claims, so it did not improve the final metrics. This negative result suggests that a zero-shot contradiction prompt is not enough; a useful conflict module would likely need stronger evidence-level NLI supervision or calibration.

We also run an oracle diagnostic by replacing retrieved development evidence with gold evidence. This raises evidence F-score to 1.0 and therefore H to 0.5806, but label accuracy falls from 0.4805 to 0.4091. The drop implies that the classifier is not simply waiting for cleaner evidence; it is sensitive to the evidence distribution seen at training and may rely on label priors or noisy retrieval patterns.

Finally, we bucket errors by whether retrieval hits at least one gold evidence passage and whether the label is correct. Only 27.9% of claims are both retrieval-hit and label-correct. Another 24.7% have a retrieval hit but a wrong label, 27.3% fail both retrieval and classification, and the remaining 20.1% are "lucky" cases where the label is correct without any gold evidence in the retrieved set, most likely from label-prior or claim-text shortcuts. This split shows that future improvements must target both candidate recall and evidence-conditioned reasoning.

5 Conclusion

Our final system is a compact, reproducible retrieval-then-classification pipeline for climate claim verification. The strongest component is retrieval: BM25 tuning, cross-encoder reranking, and relative-delta evidence selection substantially improve evidence F-score over raw BM25 top-5. The main limitation is classification. The MiniLM-MARCO classifier slightly improves over the majority-label baseline, but its strong SUPPORTS bias and zero REFUTES recall show that it has not learned robust stance reasoning.

The most promising future directions are therefore targeted rather than architectural: iterative hard-negative mining for the reranker, per-evidence NLI scoring followed by learned aggregation, and training objectives that improve minority-class recall without collapsing overall accuracy. A fuller hyperparameter search could also expose interactions missed by the sequential greedy ablation, but the current design keeps the submitted notebook feasible under the compute constraints.

Team Contributions

- Mengfan Gao (1660072): contributed to classifier experiments, ablation analysis, error analysis, and report writing.
- Jingwang Liu (1442608): contributed to data processing, baseline comparisons, reproducibility checks, and report editing.
- Kexing Ma (1697372): contributed to retrieval experiments, reranker development, evaluation, and report writing.

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